

PRD_SIH26162_FireVigil

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Product Requirements Document (PRD)

Project Name (working title): FireVigil

AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data

SIH Problem Statement ID	SIH26162
Category	Software
Theme	Disaster Management
Sponsoring Organization / Department	National Technical Research Organisation (NTRO)
Dataset Link	firms.modaps.eosdis.nasa.gov/map
Event	Smart India Hackathon 2026
Document version	2.0 (aligned to official PS text)
Owner	Ansh, MMMUT

This PRD is aligned to the **official** SIH26162 problem statement text (Background, Description, Expected Solution/Deliverables) as provided by the organization.

1. Problem Statement (Official)

Background: Industrial facilities generate thermal signatures that can be observed from space, but current satellite-based monitoring systems like NASA FIRMS cannot distinguish between different types of thermal anomalies. To address this, there is a challenge to develop an AI-enabled geospatial system that integrates thermal data, land-cover

information, industrial databases, and satellite imagery to automatically identify, classify, and monitor industrial fires and persistent thermal sources.

Description: Industrial facilities such as oil refineries, petrochemical complexes, thermal power plants, steel industries, mining areas, and LNG terminals generate thermal signatures that can be observed from space. In addition, accidental industrial fires, gas leaks, explosions, and abnormal thermal events pose significant risks to critical infrastructure, public safety, and the environment. Current satellite-based fire monitoring systems such as NASA FIRMS provide thermal anomaly detections but do not distinguish between industrial fires, gas flares, agricultural burning, mining activity, and wildfires. The challenge is to develop an AI-enabled geospatial system that can automatically identify, classify, and monitor industrial fires and persistent thermal sources by integrating thermal anomaly data, land-cover information, industrial infrastructure databases, and satellite imagery.

Expected Solution / Deliverables (official): 1. Classification and segregation of **industrial fires** from **forest fires and other natural fires**. 2. A **GIS-based solution** for data storage and visualization of the output as an **overlay over maps**.

2. Objective

Build a software system that:

1. **Ingests** near-real-time and historical thermal anomaly data from NASA FIRMS (MODIS + VIIRS) via the official dataset endpoint (firms.modaps.eosdis.nasa.gov/map).
2. **Enriches** each thermal detection with geospatial context: land-cover classification, and industrial infrastructure data (oil refineries, petrochemical complexes, thermal power plants, steel plants, mining areas, LNG terminals) sourced from OSM and/or industrial databases.
3. **Classifies** each detection — this is the core, official deliverable — into distinct categories, at minimum separating:
 - **Industrial fire** (accidental fire, gas leak, explosion, abnormal thermal event at a known industrial facility)
 - **Forest fire / other natural fire** (wildfire, non-industrial vegetation fire)
 - Additional useful sub-classes to strengthen the “distinguish between” requirement explicitly named in the PS: **gas flare** (persistent, expected industrial thermal source), **agricultural burning**, **mining activity thermal signature**
4. **Stores and visualizes** all classified thermal events in a **GIS-based system**, rendered as an interactive **overlay over maps** (the second official deliverable) — including facility context, classification label, and historical trend per location.
5. **Alerts** relevant stakeholders when a genuine, anomalous industrial fire is detected (distinct from routine/persistent thermal sources).
6. Optionally augments detections with satellite imagery (e.g., Sentinel-2/Landsat) for visual confirmation, and with land-cover datasets to strengthen the natural-vs-industrial fire distinction.

3. Goals & Success Metrics

Goal	Metric	Target (for hackathon demo / MVP)
Reduce false positives reaching operators	% of alerts flagged as “persistent/known source” vs. raw FIRMS feed	≥ 60% noise reduction
Fast detection-to-alert latency	Time from FIRMS data availability to dashboard alert	< 5 minutes (near real-time polling)
Classification accuracy	Precision/Recall on labeled validation hotspots (industrial fire vs. persistent source vs. non-industrial)	≥ 80% on demo dataset
Facility attribution coverage	% of industrial hotspots successfully matched to an OSM facility within radius	≥ 70% in pilot regions
Usability	Judges/users can locate a fire, see its classification, and view its history within 3 clicks	Pass in demo

4. Target Users / Personas



Persona	Description	Needs
Disaster Management Cell Officer (State/NDMA)	Monitors industrial hazard zones for fire emergencies	Real-time, low-noise alerts; map view; incident history
Industrial Safety / Pollution Control Analyst (CPCB/SPCB)	Tracks flaring, kiln emissions, illegal burning at facilities	Persistent-source trend reports, facility-level dashboards
NTRO / Intelligence Analyst	Needs situational awareness of thermal activity near sensitive/industrial zones	Classification confidence, satellite imagery overlay, export/API access
Hackathon Judge / Demo Viewer	Evaluates technical depth and UX	Clear, visually compelling dashboard demonstrating the full pipeline

5. Scope

5.1 In Scope (MVP for SIH demo)

- FIRMS API integration (VIIRS 375m + MODIS 1km NRT feeds) for a chosen demo region (e.g., an Indian industrial belt/state).
- OSM data ingestion (via Overpass API) for industrial `landuse=industrial`, `man_made=works`, power plants, mines, landfills, etc.
- Spatial join/matching engine (hotspot ↔ nearest facility within configurable radius).
- ML classification pipeline: rule-based + ML model using features such as recurrence frequency, brightness temperature (FRP), time-of-day pattern, spatial clustering, and facility type.
- Interactive React map dashboard (Leaflet/Mapbox) showing live + historical hotspots, filters, facility drill-down, and time-series charts per facility.
- Alerting: in-app notification feed + optional email/webhook for high-priority classified events.
- REST API for programmatic access to classified events.
- Basic auth (login) for the officer/analyst dashboard.

5.2 Stretch Goals (if time permits)

- Satellite imagery overlay (Sentinel-2 true color / thermal band snapshot at hotspot location and time) for visual verification.
- SMS/WhatsApp alert integration.
- Anomaly detection (a “persistent source” suddenly spiking beyond its historical baseline = re-flagged as high priority).
- PWA / mobile-responsive field view.
- Multi-language (Hindi/English) UI.

5.3 Out of Scope

- Physical sensor/IoT hardware deployment.
- Satellite tasking or custom satellite data acquisition.
- Legal/regulatory enforcement workflows (this is a monitoring/alerting tool, not an enforcement system).

6. Functional Requirements

ID	Requirement
FR1	System shall periodically poll/pull NASA FIRMS active fire/hotspot data (NRT) for a configured area of interest.
FR2	System shall store raw hotspot records (lat, long, timestamp, brightness/FRP, confidence, satellite/sensor source).
FR3	System shall ingest and cache OSM industrial facility geometries for the area of interest.
FR4	System shall spatially match each new hotspot to the nearest facility (if any) within a configurable distance threshold.
FR5	System shall classify each hotspot into: Industrial Fire (anomalous), Persistent Thermal Source, Non-Industrial, Unclassified/Noise.

FR6	System shall maintain a rolling historical record per facility to establish a “normal” thermal signature baseline.
FR7	System shall expose classified events via REST API (filterable by date, region, facility, class, confidence).
FR8	System shall render classified events on an interactive map with color-coded markers by class.
FR9	System shall show a facility detail view with time-series of thermal activity and past classifications.
FR10	System shall generate near-real-time alerts (in-app; email/webhook optional) for newly detected high-priority industrial fires.
FR11	System shall allow authenticated users (analysts) to manually re-label/correct a classification (feedback loop for model improvement).
FR12	System shall provide basic authentication and role-based access (Admin, Analyst, Viewer).
FR13	System shall log all ingestion, classification, and alert events for auditability.

7. Non-Functional Requirements

- **Performance:** Handle FIRMS NRT feed polling (typically updated every few hours) and process a batch of hundreds–thousands of hotspots within seconds to a few minutes.
- **Scalability:** Architecture should scale from a single-state pilot to pan-India coverage without redesign (horizontally scalable ingestion/classification workers).
- **Reliability:** Ingestion jobs must be resilient to FIRMS/OSM API downtime (retry + backoff, queue-based processing).
- **Security:** JWT-based auth, HTTPS everywhere, input validation, rate-limited public API, secrets managed via environment variables/secret manager.
- **Availability:** Target 99% uptime for the demo deployment; graceful degradation if external APIs are unavailable (serve cached/last-known data).
- **Maintainability:** Modular services (ingestion, classification, API, frontend) independently deployable and testable.
- **Data provenance:** Every classified event must retain a link back to its raw FIRMS/OSM source for auditability.

8. Data Sources

Source	Purpose	Access
NASA FIRMS (MODIS C6.1, VIIRS S-NPP/NOAA-20/NOAA-21 NRT) — official dataset link	Active fire / thermal anomaly detections	firms.modaps.eosdis.nasa.gov/map (CSV/JSON/WMS/API key)
OpenStreetMap (via Overpass API)	Industrial facility polygons/points: refineries, petrochemical complexes, power plants, steel plants, mines, LNG terminals	Free, Overpass API / OSM extracts
Land-cover dataset (e.g., ESA WorldCover, Bhuvan/ISRO LULC, Copernicus Global Land Cover)	Distinguishes forest/vegetation cover from built-up/industrial land — key input for industrial-vs-natural-fire classification	Free, satellite-derived land-cover products
Sentinel Hub / Copernicus (Sentinel-2, Sentinel-3 SLSTR) (<i>stretch</i>)	Optical/thermal imagery for visual confirmation	Free tier via Copernicus Data Space
Historical FIRMS archive	Baseline modeling for “persistent” vs “anomalous” thermal sources	Derived from FIRMS archive downloads
Administrative boundaries (India state/district GeoJSON)	Region filtering, area-of-interest scoping	Public GIS datasets (e.g., Survey of India / datameet)

9. High-Level User Flows

1. **Analyst logs in → Dashboard loads map centered on area of interest → sees color-coded hotspot markers (red = anomalous fire, orange = persistent source, grey = non-industrial/noise).**
2. **Analyst clicks a marker → side panel shows facility name (from OSM), classification, confidence score, brightness/FRP trend, and last 30 days of activity.**
3. **New high-priority fire detected → toast/notification appears in dashboard + optional email → analyst reviews and can mark as “confirmed” or “false positive” (feeds back into model).**
4. **Analyst filters by state/district/facility type/date range → exports filtered results (CSV/JSON) or views aggregated charts.**

10. Assumptions & Constraints

- FIRMS NRT data has an inherent latency (near real-time, not instantaneous) and coarse spatial resolution (375m-1km), so classification is probabilistic, not forensic proof.
- OSM completeness for industrial facilities varies by region; matching accuracy depends on OSM data quality in the demo area.
- For the hackathon MVP, classification will likely use a hybrid of **rule-based heuristics** (recurrence count, day/night pattern, distance-to-facility) plus a **lightweight ML model** trained on a labeled subset, rather than a large deep-learning model, given the 36-hour build constraint.
- Team will use **Node.js + Express.js + React** as mandated, with a **Python microservice** for the ML/geospatial-heavy classification logic (justified in the architecture document).

11. Milestones (Hackathon-oriented plan)

Phase	Deliverable
Day 0 (prep)	FIRMS API key, OSM extract for demo region, repo scaffolding, wireframes
Hour 0-8	Backend skeleton (Express APIs), DB schema, FIRMS ingestion job
Hour 8-16	OSM matching engine, classification service (rules + baseline ML), React map dashboard shell
Hour 16-26	Classification model refinement, alerting, facility detail views, charts
Hour 26-32	Auth, polish UI/UX, seed demo data, edge-case handling
Hour 32-36	Testing, deployment, pitch deck, demo rehearsal

12. Confirmed Scope Notes (from official PS)

- **Industrial sectors explicitly named:** oil refineries, petrochemical complexes, thermal power plants, steel industries, mining areas, LNG terminals.
- **Thermal event types explicitly named:** accidental industrial fires, gas leaks, explosions, abnormal thermal events, gas flares.
- **Confusable categories to distinguish against:** forest fires/wildfires, agricultural burning, mining activity.
- **Official dataset:** firms.modaps.eosdis.nasa.gov/map — no additional dataset was provided beyond this, so OSM, land-cover, and any industrial facility registries are to be sourced by the team.
- **The two deliverables judges will grade against are explicit and narrow:** (1) industrial-vs-natural fire classification, and (2) a GIS map-overlay visualization. Everything else in this PRD (alerting, feedback loop, baselines) is value-add on top of that core, not a substitute for it — prioritize (1) and (2) first if time is short.